

SQL Interview Terms – Quick Notes

Assume we have an Employee table:

```
Employee(emp_id, name, salary)
```

1. Schema

Interview answer: A schema is the logical structure or blueprint of a database. It defines how tables, columns, relationships, constraints, and other database objects are organized.

Easy meaning: Schema = blueprint of the database.

2. Table

Interview answer: A table is a collection of related data organized into rows and columns. It is used to store data in a relational database.

3. Row

Interview answer: A row represents a single entry or instance of data in a table.

4. Record

Interview answer: A record is a complete set of related data representing one entity or instance. In a relational table, a record is generally represented by a row.

Remember: Row and record are generally used interchangeably in SQL.

5. Column

Interview answer: A column represents a particular attribute or property of the data stored in a table. Values in a column generally have the same data type.

6. Field

Interview answer: A field is a single data value at the intersection of a row and a column.

Example: In 101 | Rahul | 50000, the value Rahul is a field value.

7. Attribute

Interview answer: An attribute is a property or characteristic that describes an entity. In a relational database, an attribute is usually represented as a column.

Distinction: Attribute is the conceptual/design term; column is the relational database implementation.

8. Entity

Interview answer: An entity is a real-world object or concept about which we want to store information in a database.

Examples: Employee, Student, Customer, Product, Department.

9. Relationship

Interview answer: A relationship defines how two or more entities are associated with each other.

Common types: One-to-One (1:1), One-to-Many (1:N), Many-to-Many (M:N).

Example: Department 1 → many Employees.

10. Metadata

Interview answer: Metadata is data that describes the structure or properties of other data.

Examples: table name, column names, data types, constraints.

Easy meaning: Metadata = data about data.

11. Query

Interview answer: A query is a request made to a database to retrieve or manipulate data.

Example:

```
SELECT name FROM Employee WHERE salary > 50000;
```

12. RDBMS

Interview answer: RDBMS stands for Relational Database Management System. It is software used to store, manage, and manipulate data in relational tables, where tables can be related using keys.

Examples: MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server.

Quick Revision

Topic	Interview Definition
Schema	Blueprint/structure of a database
Table	Collection of related data in rows and columns
Row	Single entry in a table
Record	Complete set of data representing one instance
Column	Represents a particular property/attribute
Field	Single value at a row-column intersection
Attribute	Property/characteristic of an entity
Entity	Real-world object about which data is stored
Relationship	Association between entities
Metadata	Data that describes other data
Query	Request to retrieve or manipulate database data
RDBMS	Software that manages relational databases

Interview focus: Be ready to explain row vs record, column vs field, attribute vs column, entity vs table, schema vs database, and DBMS vs RDBMS.